WASP-2b

R-0273

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-2_230816` · created 2026-07-17T13:51:26+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460172.703 +/- 0.0076 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.047
Transit depth (Rp/Rs)^2	2.57 +/- 1.51 [%]
Orbital Inclination (inc)	87.9 +/- 2.9
Airmass coefficient 1 (a1)	2.31 +/- 0.7
Airmass coefficient 2 (a2)	-0.67 +/- 0.37
Scatter in the residuals of the lightcurve fit is	29.32 %
Best Comparison Star	#3 - [103, 82]
Optimal Aperture	1.75
Optimal Annulus	11.1
Transit Duration (day)	0.089 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.16 ± 0.047 vs 0.1326 (deviation 0.39σ)
Duration fit vs published	0.089 d vs 0.0737 d (deviation 0.57σ)
Mid-time O–C	8.8 min
Depth SNR	2.3
Residual scatter	29.32 %

Light curve

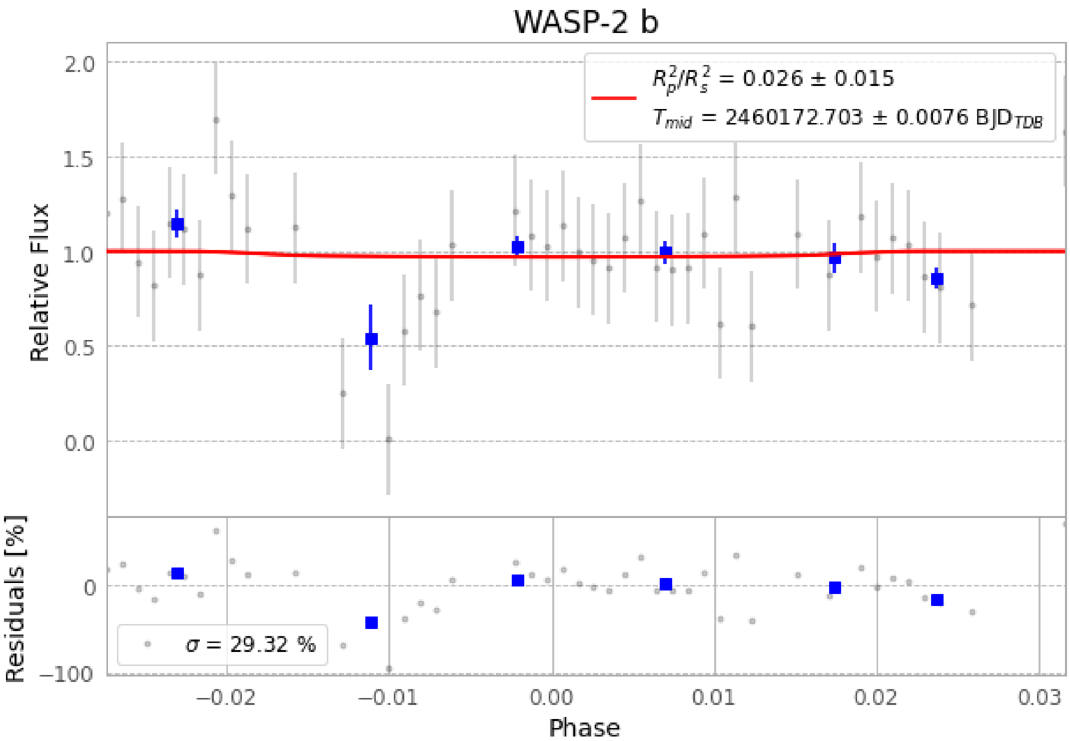

Figure 1 — FinalLightCurve_WASP-2 b_2023-08-15.png (SHA-256 759c9ef2c0935152…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [282, 212], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 116642036b087c5d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 5cfc163`

Data provenance

65 FITS frames (43 MB), `WASP-2230816031815.FITS` → `WASP-2230816063015.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-230816.log (3e729b2ed1df…), FinalLightCurve_WASP-2 b_2023-08-15.png (759c9ef2c093…), FinalLightCurve_WASP-2 b_2023-08-15.csv (305bfdc1e061…)

Compute resources

Wall time 205.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)